Image Geolocation with Geographic Classification and Spherical Residual Refinement

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Abstract

This project investigates image geolocation from street-level photographs by combining geographic classification with local coordinate refinement. Training locations are grouped into 128 geographic clusters using a 3D unit sphere representation. The proposed approach combines coarse geographic classification with local spherical residual refinement. A MobileNetV3 Large model, initialized with ImageNet pretrained weights is trained with an auxiliary country label. A spherical residual head refines the predicted cluster center. On the held out validation set, direct coordinate regression achieved a median validation error of 284.79 km, the K=128 geographic clustering with country supervision reduced it to 102.24 km. The final spherical residual refinement further reduced the median error to 97.95 km.

1 Approach

The initial approach treated image geolocation as direct coordinate regression. However, predicting precise coordinates from visually ambiguous street scenes proved difficult.

The task was then reformulated as geographic classification followed by local coordinate refinement. The model first predicts a coarse geographic region and then applies a learned spherical residual to refine the predicted location.

2 Architecture

The final model uses a shared MobileNetV3-Large backbone with cluster, country, and spherical residual heads, as shown in Figure 1.

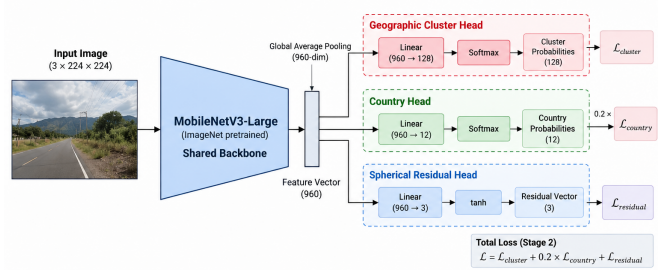


Figure 1: Training Pipeline

The predicted cluster center is refined using the spherical residual and converted to latitude and longitude, as shown in Figure 2.

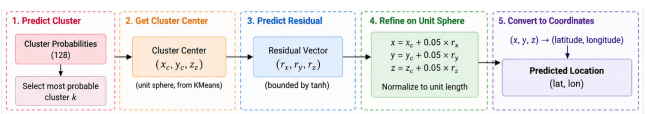


Figure 2: Inference pipeline.

2.1 Geographic representation

Applying KMeans directly to raw latitude and longitude is problematic because geographic coordinates do not form a Euclidean plane. Therefore, each location is first mapped on to the 3D unit sphere,

$$x = \cos\phi\cos\lambda, \quad y = \cos\phi\sin\lambda, \quad z = \sin\phi,$$

where ϕ and λ are latitude and longitude. The clustering is performed only on the training portion of the held-out split. Each image receives the ID of its corresponding geographic cluster. The strongest configuration uses K = 128 geographic clusters. The shared MobileNetV3-Large backbone feeds a 128-class cluster head and a 12-class country head. The training objective for this stage:

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{cluster}} + 0.2\,\mathcal{L}_{\text{country}}.$$

The country prediction is used as auxiliary supervision. The geographic localization is primarily determined by the cluster prediction.

2.2 Spherical Residual Refinement

A three-dimenstional residual head was added to predict a local correction around the cluster center. Because using only a cluster center introduces quantization error, as every image assigned to the same cluster receives the same representative location.

For true unit-sphere coordinate $\mathbf{x}$ and assigned cluster center $\mathbf{c}$, the residual training target is

$$\mathbf{r}^* = \frac{\mathbf{x} - \mathbf{c}}{0.05}.$$

The residual head outputs three values bounded using tanh. At inference,

$$\hat{\mathbf{x}} = \mathbf{c}_{\hat{k}} + 0.05\,\hat{\mathbf{r}},$$

where $\mathbf{c}_{\hat{k}}$ is the center of the predicted cluster. The resulting vector is normalized back to unit length before conversion to latitude and longitude.

The residual head is zero-initialized. The backbone, cluster head, and country head are initialized from the trained K=128 + country checkpoint. The complete network is then jointly fine-tuned with

$$\mathcal{L} = \mathcal{L}_{\text{cluster}} + 0.2\,\mathcal{L}_{\text{country}} + \mathcal{L}_{\text{residual}},$$

where the residual term is Smooth L1 loss.

3 Experiments and Design Decision

A country-stratified 80/20 split with random seed 42 was used for model development, producing 9,406 training images and 2,352 validation images. The hidden public holdout was not used for model selection.

3.1 Direct coordinate regression

The first approach predicted normalized latitude and longitude using a MobileNet backbone. With ImageNet initialization, the best regression model reached a validation median error of 284.79km, with 37.97% of predictions within 200km and 81.89% within 750km.

The observation was the relatively high <750km rate, suggesting that the network could often infer the approximate geographic area, but direct regression was not sufficiently precise.

Method	Med.↓	Mean↓	<200↑	<750↑
Regression	284.79	467.64	37.97	81.89
K=80 + country	126.19	459.76	60.12	80.27
K=96 + country	122.97	479.87	59.48	78.61
K=128 cluster	105.11	485.10	60.03	78.27
K=128 + country	102.24	**463.24**	**61.22**	**79.21**
K=128 + country + res.	**97.95**	467.06	60.88	78.57

Table 1: Held-out validation results. Distances are in km and threshold columns are percentages.

3.2 Geographic cluster classification

The next approach converted the continuous geolocation problem into geographic classification. The median error decreased steadily as cluster resolution increased, with K=128, cluster classification alone reached a 105.11 km median.

However, increasing the number of clusters reduces the number of training examples available per class. It's a trade off between number of clusters and better geographic precision. Among the tested configurations, K = 128 provided the strongest validation performance.

3.3 Auxiliary country supervision

The provided country labels give a natural coarse geographic hierarchy. A second prediction head was therefore introduced to classify the image into one of the 12 countries.

Adding country supervision with weight 0.2 reduced the median error from 105.11 km to 102.24 km and increased the <200km rate from 60.03% to 61.22%. The mean error also improved from 485.10 km to 463.24 km, suggesting that country supervision helped with some larger errors.

3.4 Spherical residual refinement

Initial residual experiments did not improve the cluster model. The final approach instead starts from the trained K=128 + country checkpoint. The residual target is scaled by 0.05, the prediction is bounded with tanh, and the residual head is zero-initialized. Joint fine-tuning at a lower learning rate reduced the median error from 102.24 km to 97.95 km, an improvement of approximately 4.2%.

4 Failed Approaches and Failure Analysis

Soft cluster decoding which combines multiple cluster centers using their predicted probabilities, performed worse than taking the most probable cluster. One possible reason is that similar scenes can correspond to distant locations. Hence averaging predictions can cause incorrect intermediate location.

Initial residual experiments attempted to predict a continuous correction from the geographic cluster center but did not improve performance. These experiments showed that simply adding a regression objective was insufficient for learning useful local corrections.

There are two additional evaluations that used to determine whether the improvement actually came from the residual correction.

Training only the residual head did not improve the median error. However, for the jointly fine-tuned checkpoint, applying the residual improved the median from 102.51 km to 97.95 km, confirming that the residual provides useful local refinement.

Method	Med.↓	Mean↓	<200↑	<750↑
Baseline	102.24	463.24	61.22	79.21
Frozen + residual	103.41	**462.71**	**61.82**	**79.34**
Joint residual	**97.95**	467.06	60.88	78.57

Table 2: Residual training ablation.

Also, most large errors were caused by cross-country confusion. This suggests that visually similar scenes across different countries remain difficult to distinguish.

5 Final model and Training Pipeline

For the final training run, KMeans is fitted using all 11,758 provided labeled locations to create 128 geographic clusters. Training then proceeds in two stages.

Stage 1 trains the MobileNetV3-Large cluster–country model for 40 epochs using batch size 64, learning rate 10^{-3}, weight decay 10^{-3}, country-loss weight 0.2, and label smoothing 0.1.

Stage 2 adds the zero-initialized residual head and initializes all existing weights from the Stage 1 checkpoint. The complete model is jointly fine-tuned for six epochs using learning rate 10^{-4}, weight decay 10^{-3}, country-loss weight 0.2, and residual-loss weight 1.0. Epoch 6 was selected by validation median among the 10 development epochs. This may slightly overestimate the reported improvement.

Property	**Final configuration**
Backbone	MobileNetV3-Large
Geographic clusters	128
Country classes	12
Residual dimensions	3
Residual scale	0.05
Residual activation	tanh
Stage 1 epochs	40
Stage 2 epochs	6
Stage 1 learning rate	0.001
Stage 2 learning rate	0.0001
Batch size	64
Final parameters	4,385,215

Table 3: Final model configuration.

The Stage 1 network contains 4,381,372 parameters. The residual head adds 3,843 parameters, giving 4,385,215 parameters in the complete final network. The final Stage 2 checkpoint contains the complete network, and only this model is loaded during final inference. The final model is applied to all 2,400 unlabeled public holdout images to generate the required predictions.csv.

Conclusion

The most important improvement came from reformulating geolocation as geographic classification. This final refinement reduced the median error to 97.95 km, the best validation median obtained during model development, while keeping the complete model at only 4.39 million parameters.

The classification-based formulation builds on prior work discretizing geographic space for image geolocation [1].

References

[1] T. Weyand, I. Kostrikov, and J. Philbin, "PlaNet: Photo Geolocation with Convolutional Neural Networks," *Proc. ECCV*, pp. 37–55, 2016.